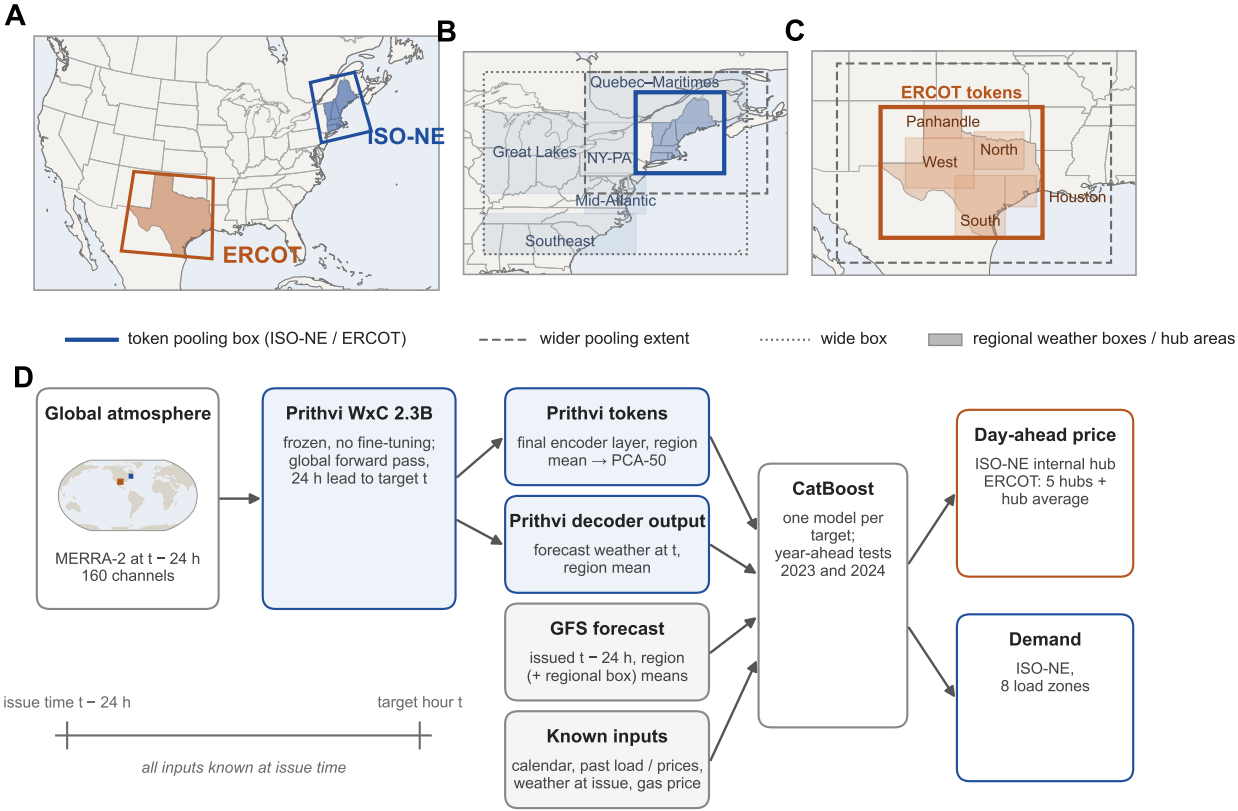

A
ISO-NE
ERCOT
B
Quebec–Maritimes
Great Lakes
NY-PA
Mid-Atlantic
Southeast
C
ERCOT tokens
Panhandle
West
North
Houston
South
token pooling box (ISO-NE / ERCOT)
wider pooling extent
wide box
regional weather boxes / hub areas
D
Global atmosphere
MERRA-2 at t − 24 h
160 channels
Prithvi WxC 2.3B
frozen, no fine-tuning;
global forward pass,
24 h lead to target t
Prithvi tokens
final encoder layer, region mean → PCA-50
Prithvi decoder output
forecast weather at t, region mean
GFS forecast
issued t − 24 h, region (+ regional box) means
Known inputs
calendar, past load / prices, weather at issue, gas price
CatBoost
one model per target;
year-ahead tests
2023 and 2024
Day-ahead price
ISO-NE internal hub
ERCOT: 5 hubs + hub average
Demand
ISO-NE,
8 load zones
issue time t − 24 h
target hour t
all inputs known at issue time